Experiment 28: K-Means Clustering for Customer Segmentation

Aim: To group customers into different segments using K-Means clustering.

Algorithm:

1. Import K-Means.
2. Create customer shopping data.
3. Train the K-Means model.
4. Get new customer details.
5. Predict the customer segment.

Code:

```
from sklearn.cluster import KMeans

X = [[10, 100], [15, 150], [20, 200],
    [80, 800], [90, 900], [100, 1000]]

model = KMeans(n_clusters=2, random_state=42, n_init=10)
model.fit(X)

amount = float(input("Enter purchase amount: "))
visits = float(input("Enter shopping frequency: "))

result = model.predict([[amount, visits]])

print("Customer Segment:", result[0])
```

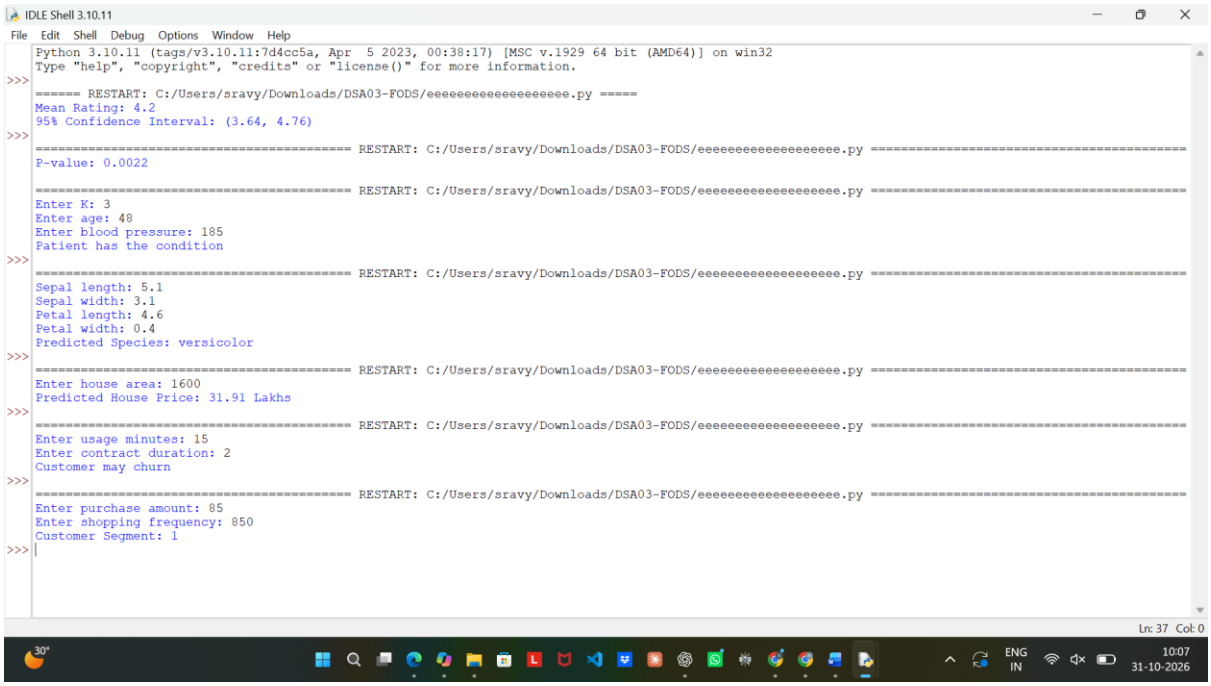
Input:

Enter purchase amount: 85

Enter shopping frequency: 850

Output:

Customer Segment: 1



Result: Customers were successfully grouped into segments using K-Means clustering.